

E2-Conditioned Finite-Horizon Effective Capacity for Public-Safety MCX over Shared O-RAN

Jingqing Wang, *Member, IEEE* and Wenchi Cheng, *Senior Member, IEEE*

Abstract—Supporting public-safety Mission Critical Services (MCX) over a shared Open radio access network (O-RAN) requires service assurance over finite incident horizons, while ordinary mobile traffic competes for the same resources and heterogeneous E2 domains expose different observations, control actions, and actuation latencies. Existing RAN key performance indicators are retrospective, whereas conventional effective capacity characterizes an asymptotic stationary regime and therefore suppresses both the initial E2-observed condition and the finite selection-to-actuation transient. To solve this problem, in this paper we develop an E2-conditioned finite-horizon effective capacity (FH-EC) framework for public-safety MCX over shared O-RAN. Specifically, we first establish a finite-horizon O-RAN MCX-based service model that incorporates E2-observed network states, control actuation latency, and correlation-aware connectivity diversity. Based on this model, we derive an FH-EC formulation that characterizes the executable service capability within a finite mission horizon. Furthermore, we transform FH-EC into confidence-calibrated capability profiles and develop an MCX orchestration framework with contract certification, shared-resource protection, and FH-EC-driven profile selection at the Near-RT RAN Intelligent Controller (RIC). Coupled MATLAB/ns-3 evaluations demonstrate the predicted short-horizon state and actuation effects and show that adaptive connectivity selection improves MCX supportability under O-DU degradation while satisfying the configured multi-QoS and non-MCX protection requirements.

Index Terms—Finite-horizon effective capacity, multi-QoS provisioning, E2 exposure, HRLLC, public-safety MCX, O-RAN.

I. INTRODUCTION

OPEN radio access network (O-RAN) has emerged as a promising architecture for next-generation networks by introducing programmable control, open interfaces, and AI-driven network orchestration [1]. Through the Service Management and Orchestration (SMO) framework, the Non-Real-Time RAN Intelligent Controller (Non-RT RIC), the Near-Real-Time RIC (Near-RT RIC), and standardized interfaces, network intelligence can be deployed across disaggregated O-RAN Central Unit (O-CU) and O-RAN Distributed Unit (O-DU) entities operating under heterogeneous implementations and administrative domains [2]. This architectural evolution creates an important opportunity for supporting public-safety Mission Critical Services (MCX) over shared mobile infrastructure [3, 4]. However, programmability alone does

not provide service assurance. Open interfaces make a RAN configuration observable and controllable, but not necessarily certifiably supportable for a critical service.

This distinction is particularly important for MCX. Control and group-voice bearers require bounded delivery delay and a low probability of terminal non-delivery. Real-time video is additionally sensitive to packet-delay variation while status-update traffic requires information freshness. Finite-blocklength transmission further couples payload rate, blocklength, and reliability [5]. Consequently, delivered rate, delay, jitter, age of information (AoI) [6, 7], and reliability represent different, service-dependent quality-of-service (QoS) dimensions rather than interchangeable manifestations of average throughput [8–10]. Their interactions are central to hyper-reliable and hyper-low-latency communications (HRLLC) [11–13]. In a shared RAN, these requirements are coupled to ordinary mobile services. Reserving additional resources may improve MCX supportability, but can substantially reduce non-MCX utility or starve a protected subset of ordinary users. An MCX assurance mechanism must therefore certify service-specific risks while explicitly constraining the collateral degradation imposed on non-MCX traffic.

Although O-RAN provides the architectural mechanisms needed to act on this problem, it introduces a capability-semantic gap. E2 measurements describe what an E2 node reports, while its advertised RAN functions specify what the Near-RT RIC may observe and control. Different E2 nodes may expose different measurement granularity, reporting periods, control actions, and actuation delays. These raw observations and action sets do not directly quantify the service that an executable configuration can support. At the admission epoch of an MCX flow, the Near-RT RIC must select a profile specifying the connectivity mode, participating domains, local execution policy, and reserved resources. The central challenge, therefore, is to determine the level of service a candidate profile can guarantee within a finite MCX mission window [14], accounting for real-time E2 telemetry and selection-actuation delays. Addressing this requires a unified capability coordinate system through which heterogeneous E2 configurations can be compared and safely selected.

The mission window in this question is not merely the periodic execution interval of the Near-RT RIC. It is the finite commitment window over which an admitted MCX profile and its service certificate must remain valid. After admission, slot-level scheduling, hybrid automatic repeat request (HARQ), link adaptation, and other local decisions may continue to adapt within the selected O-CU/O-DU policy, but the certified profile cannot be freely replaced without invalidating the capa-

This work of Jingqing Wang and Wenchi Cheng was supported in part by the National Natural Science Foundation of China under Grant 62341132 and the Natural Science Basic Research Program of Shaanxi under Grant 2024JC-YBQN-0642.

Jingqing Wang and Wenchi Cheng are with the State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an, China (e-mails: jqwangxd@xidian.edu.cn; wccheng@xidian.edu.cn).

bility. Moreover, profile-selection runtime, E2 control-message delivery, processing, and distributed O-CU/O-DU actuation consume a non-negligible portion of a short mission window. During this transient, service is governed by the pre-actuation policy rather than the newly selected one. The service available over the complete mission window therefore depends jointly on the E2-conditioned initial state, the finite control transient, and the post-actuation execution policy.

Existing RAN key performance indicators cannot directly answer this finite-horizon question because they are primarily descriptive rather than forward-looking and action-conditioned. From a theoretical perspective, effective capacity provides a risk-sensitive abstraction for statistical delay provisioning [15, 16]. However, conventional effective capacity is defined through the asymptotic log-moment rate of a stationary and ergodic service process. Its steady-state limit suppresses dependence on the initial E2-observed condition and assigns a vanishing contribution to a finite control transient. This approximation is appropriate when the operational horizon is much longer than the channel mixing time, congestion-burst duration, mobility interval, infrastructure-recovery time, and control-actuation delay. It can, however, overestimate or underestimate the service available when the MCX commitment window is comparable to these timescales. The required abstraction should therefore retain the risk-sensitive interpretation of effective capacity over the actual mission horizon, explicitly incorporate the selection-and-actuation transient, and recover conventional effective capacity as the horizon grows.

Existing studies provide complementary components of such an abstraction. Finite-horizon scheduling has been investigated for wireless networked control systems [14], while finite-blocklength and mission-effective-capacity formulations characterize communication reliability and dependability [17]. In O-RAN, resource allocation has been studied under network-slicing constraints [18], and stochastic network calculus has been used to support latency-tail-sensitive orchestration [19]. CoO-RAN and ns-O-RAN further provide valuable foundations for learning-based xApps, closed-loop control, and system-level O-RAN experimentation [20, 21]. Packet duplication and multi-connectivity can improve delivery reliability by exploiting spatial diversity, although their benefit depends on resource consumption and correlated radio or infrastructure failures [22]. These studies establish the foundations for programmable control, statistical QoS provisioning, and connectivity diversity. However, they do not jointly provide an E2-conditioned and actuation-aware finite-horizon capability contract that carries simultaneous finite-sample service and multi-QoS certificates and can be consumed by a Near-RT RIC while protecting ordinary traffic in a shared O-RAN.

To bridge this gap, we develop an E2-conditioned finite-horizon effective-capacity (FH-EC) framework for public-safety MCX over shared O-RAN. Specifically, first, we develop an E2-conditioned and actuation-aware finite-horizon service model for MCX over shared O-RAN. Second, we establish the long-horizon consistency of the proposed FH-EC under a finite E2-control transient and a primitive post-actuation Markov-additive kernel. Third, we develop a contract-preserving O-RAN profile-orchestration framework.

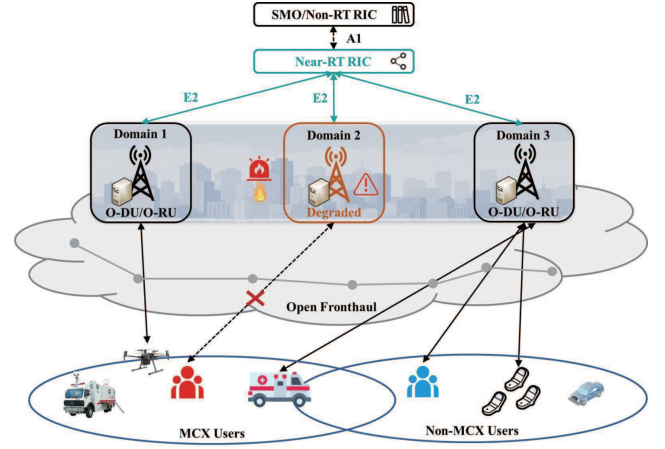


Fig. 1. Shared Open 6G RAN architecture for E2-conditioned FH-EC service profile orchestration.

Coupled MATLAB/ns-3 system-level evaluation results evaluate the capability model and assess the effects of the proposed E2-conditioned finite-horizon multi-QoS orchestration.

The rest of this paper is organized as follows. Section II builds the O-RAN MCX service model. Section III derives E2-conditioned FH-EC expressions. Section IV establishes the FH-EC-driven MCX profile orchestration in O-RAN. Section V validates and assesses the system performance. Section VI concludes the paper.

II. O-RAN MCX SERVICE MODEL

A. Network Architecture and Information Model

As illustrated in Fig. 1, the considered O-RAN architecture comprises an SMO/Non-RT RIC, a logical Near-Real-Time RIC coordination plane, one or more O-CU-CP/O-CU-UP entities, and multiple O-DU/O-RAN Radio Unit (O-RU) nodes. The O-CU-UP hosts the user-plane Packet Data Convergence Protocol (PDCP) functions, while Radio link Control (RLC), MAC, HARQ, and slot-level scheduling are executed at the O-DUs [23, 24].

Let $\mathcal{I} = \{1, \dots, I\}$ be the logical E2 control domains. Domain $i \in \mathcal{I}$ contains the O-DU/E2-node set, denoted by $\mathcal{B}_i$, where $\mathcal{B} = \cup_{i \in \mathcal{I}} \mathcal{B}_i$ and $\mathcal{B}_i \cap \mathcal{B}_j = \emptyset, i \neq j$, and $\mathcal{B}$ is the set of O-DUs. Different E2 nodes may expose different measurements, control functions, and actuation delays. We characterize the E2 exposure profile, denoted by Ξ_i , of domain i by

$$\Xi_i \triangleq (\mathbf{h}_i^{\text{KPM}}, \mathcal{A}_i^{\text{RC}}, T_i^{\text{KPM}}, \tau_{i,l}^{\text{act}}) \quad (1)$$

where $\mathbf{h}_i^{\text{KPM}}$ is the domain-specific Key Performance Measurement (KPM) observation and aggregation map, $\mathcal{A}_i^{\text{RC}}$ is the set of RAN-control commands exposed to the orchestrator, T_i^{KPM} is the KPM reporting period, and $\tau_{i,l}^{\text{act}}$ is the delay from issuing a control command to its effective application. Let T_{sl} and T_x denote the O-DU scheduling-slot duration and the Near-RT RIC orchestration interval, respectively with $T_{sl} \ll T_x$. We define $t_l = lT_x$ as the beginning of Near-RT orchestration epoch l , and let $k_l = \lfloor t_l/T_{sl} \rfloor$ as the corresponding slot index.

The observation delay, denoted by $\delta_{i,l}^{\text{obs}}$, measured in slots is given as follows:

$$\delta_{i,l}^{\text{obs}} = \lceil \frac{\tau_{i,l}^{\text{obs}}}{T_{sl}} \rceil. \quad (2)$$

Let $z_i[k]$ denote the physical state of domain i , and let h_i^{KPM} be its E2SM-KPM observation and aggregation map. Then, the latest E2-exposed observation, denoted by $\hat{z}_i[l]$, available at t_l is modeled as:

$$\hat{z}_i[l] = h_i^{\text{KPM}}(z_i[k_l - \delta_{i,l}^{\text{obs}}]) + \nu_i[l] \quad (3)$$

where $\nu_i[l]$ represents measurement noise, time/frequency aggregation error, quantization, compression, and unreported-state uncertainty. The information available to the orchestrator, denoted by $\mathcal{F}_l^{\text{E2}}$, consists of the time-stamped E2 observations and past exposed control outcomes.

An MCX flow s is admitted at orchestration epoch l_s , whose beginning time is $t_{l_s} = t_s^0$. At admission, the Near-RT RIC selects one persistent profile a , which is given as follows:

$$a = (m_a, \mathcal{I}_a, \varpi_{s,a}, \mathbf{b}_{s,a}, \tau_{s,a}^{\text{act}}), \quad (4)$$

where $m_a \in \{\text{SC}, \text{DP}\}$ is the connectivity mode, $\mathcal{I}_a$ is the participating E2-domain set, $\varpi_{s,a}$ is the fixed O-CU/O-DU execution-policy tuple, $\mathbf{b}_{s,a}$ is the reserved-resource vector, and $\tau_{s,a}^{\text{act}}$ is the profile-actuation delay. Once admitted, the profile identifier and its resource reservation remain unchanged over a finite operational window, denoted by $\mathcal{T}_s$, given by

$$\mathcal{T}_s = [t_s^0, t_s^0 + \Delta_s) \quad (5)$$

where Δ_s is the mission horizon, which is given by

$$\Delta_s = K_s \mathcal{T}_{sl}, \quad K_s \in \mathbb{Z}_+ \quad (6)$$

Slot-level scheduling, HARQ, and link adaptation may still adapt to local observations according to the fixed local policy contained in $\varpi_{s,a}$.

Let $T_{\text{sel}}[l_s]$ denote the runtime required by the Near-RT RIC to complete profile selection for flow s at its admission epoch l_s . The runtime is upper bounded as $0 \leq T_{\text{sel}}[l_s] \leq T_{\text{sel}}^{\text{max}}$. For candidate profile a , the total control-to-effect delay, denoted by $\tau_{s,a}^{\text{ctl}}$, is defined as follows:

$$\tau_{s,a}^{\text{ctl}} \triangleq T_{\text{sel}}^{\text{max}} + \tau_{s,a}^{\text{act}}, \quad (7)$$

where $\tau_{s,a}^{\text{act}}$ includes the E2 control-message delivery, processing, and distributed O-CU/O-DU actuation delay. The corresponding number of pre-actuation slots and the effective slot of profile a are, respectively, given as follows:

$$\begin{cases} d_{s,a}^{\text{ctl}} \triangleq \left\lceil \frac{\tau_{s,a}^{\text{ctl}}}{T_{sl}} \right\rceil; \\ k_{s,a}^{\text{eff}} \triangleq k_{l_s} + d_{s,a}^{\text{ctl}}. \end{cases} \quad (8)$$

Only profiles satisfying $d_{s,a}^{\text{ctl}} < K_s$ are eligible for admission. During the interval $[k_{l_s}, k_{s,a}^{\text{eff}})$, the target profile has not yet become effective. We denote the corresponding pre-actuation policy by ϖ_s^{pre} . For a newly admitted MCX flow, ϖ_s^{pre} provides no guaranteed payload service, whereas for an explicitly supported reconfiguration it may represent the incumbent policy. From slot $k_{s,a}^{\text{eff}}$ onward, the selected policy $\varpi_{s,a}$ remains unchanged until the end of the mission window.

B. Finite-Horizon SC/DP Radio Service

Let $\mathcal{S} = \mathcal{S}_M \cup \mathcal{S}_N$ and $\mathcal{S}_M \cap \mathcal{S}_N = \emptyset$, where $\mathcal{S}_M$ and $\mathcal{S}_N$ denote the MCX and non-MCX flow sets. An MCX bearer may represent a control stream, a video stream, or a data/status-update stream [25].

Packet n of flow s arrives at the RAN ingress at time $g_{s,n}$ and contains $L_{s,n}$ information bits. The number of arriving bits, denoted by $A_s[k]$, in slot k is given as follows:

$$A_s[k] = \sum_{n: g_{s,n} \in [kT_{sl}, (k+1)T_{sl})} L_{s,n}. \quad (9)$$

Let $\chi_b^W[k] \in [0, 1]$ and $\chi_b^P[k] \in [0, 1]$ denote the available bandwidth and power fractions of O-DU b . Let $\omega_{b,f}[k]$ and $p_{b,f}[k]$ denote the bandwidth and transmit power allocated by O-DU b to flow $f \in \mathcal{S}$ in slot k . The per-slot constraints are listed as follows:

$$\begin{cases} \sum_{f \in \mathcal{S}} \omega_{b,f}[k] \leq \chi_b^W[k] W_b; \\ \sum_{f \in \mathcal{S}} p_{b,f}[k] \leq \chi_b^P[k] P_b^{\text{max}}, \end{cases} \quad (10)$$

where W_b and P_b^{max} are the nominal bandwidth and maximum transmit power of O-DU b , respectively. After accounting for pilot and control overhead, the number of channel uses, denoted by $n_{b,f}[k]$, assigned to flow f is modeled as follows:

$$n_{b,f}[k] = \lfloor \eta_b \omega_{b,f}[k] T_{sl} \rfloor \quad (11)$$

where $\eta_b \in (0, 1)$ is the usable time-frequency fraction. For $\omega_{b,f}[k] > 0$, the effective SINR, denoted by $\gamma_{b,f}[k]$, is modeled as follows:

$$\gamma_{b,f}[k] = \frac{p_{b,f}[k] g_{b,f}[k]}{N_0 \omega_{b,f}[k] + I_{b,f}[k]} \quad (12)$$

where $g_{b,f}[k]$ is the effective channel power gain, N_0 is the noise power spectral density, and $I_{b,f}[k]$ is the aggregate co-channel interference. Assuming quasi-static channel conditions over one codeword, under the finite-blocklength normal approximation, the achievable coding rate, denoted by $R_{b,f}[k]$, in bits per channel use is modeled as:

$$R_{b,f}[k] \approx \left[\log_2(1 + \gamma_{b,f}[k]) - \sqrt{\frac{V(\gamma_{b,f}[k])}{n_{b,f}[k]}} Q^{-1}(\epsilon_{b,f}[k]) \right]^+ \quad (13)$$

where $\epsilon_{b,f}[k]$ denotes the target decoding error probability, $Q^{-1}(\cdot)$ is the inverse Gaussian Q-function, and $V(\gamma_{b,f}[k])$ is the channel dispersion, which is defined as follows:

$$V(\gamma_{b,f}[k]) = \left[1 - \frac{1}{(1 + \gamma_{b,f}[k])^2} \right] (\log_2 e)^2. \quad (14)$$

Define the scheduled information payload, denoted by $L_{b,f}[k]$, as follows:

$$L_{b,f}[k] = n_{b,f}[k] R_{b,f}[k] \quad (15)$$

The decoding-success indicator is defined as $\xi_{b,f}[k] \in \{0, 1\}$, with

$$\Pr\{\xi_{b,f}[k] = 1 | \gamma_{b,f}[k], n_{b,f}[k]\} = 1 - \epsilon_{b,f}[k]. \quad (16)$$

Therefore, we denote the successfully decoded leg-level service, denoted by $\tilde{S}_{b,f}[k]$, as follows:

$$\tilde{S}_{b,f}[k] = \xi_{b,f}[k] L_{b,f}[k]. \quad (17)$$

The lower-layer queue, RLC, and HARQ processes map $\tilde{S}_{b,s}[k]$ to packet-completion times.

Each MCX bearer is anchored at one transmitting-side O-CU-UP/PDCP entity. Let T_s^{out} denote the terminal-outcome timeout. Let $d_{b,s,n,u}^{\text{link}}$ be the completion time of packet n at receiver u through O-DU b , as induced by the lower-layer queue/HARQ process. Let $\mathcal{C}_{s,n}(a)$ be the O-DU set used by candidate a . It is a singleton for SC and contains at least two O-DUs for DP. The completion time, denoted by $d_{s,n,u}^a$, at receiver u is given as follows:

$$d_{s,n,u}^a = \min_{b \in \mathcal{C}_{s,n}(a)} d_{b,s,n,u}^{\text{link}} \quad (18)$$

Then, the group-completion time, denoted by $d_{s,n}^a$, is given as follows:

$$d_{s,n}^a = \inf \left\{ t : \frac{1}{|\mathcal{U}_s|} \sum_{u \in \mathcal{U}_s} \mathbf{1} \{ d_{s,n,u}^a \leq t \} \geq \zeta_s \right\}, \quad (19)$$

where ζ_s is the required receiver-completion fraction and $\mathcal{U}_s$ is the target receiver set. The bearer-level group-completion indicator, denoted by $Y_{s,n}^a$, is defined as follows:

$$Y_{s,n}^a = \mathbf{1} \{ d_{s,n}^a \leq g_{s,n} + T_s^{\text{out}} \}. \quad (20)$$

Thus, the unique-payload service is given by

$$S_{s,a}[k] = \sum_n L_{s,n} \mathbf{1} \{ d_{s,n}^a \in [kT_{\text{sl}}, (k+1)T_{\text{sl}}) \}. \quad (21)$$

DP consumes resources on every selected leg but contributes each successfully delivered payload only once. Thus, a duplicated packet consumes resources on all selected links but contributes its payload to $S_{s,a}[k]$ only once. The first successfully received copy determines the packet delivery time, which is also consistent with standard packet-delay-variation measurement practice.

C. Finite-Horizon Multi-QoS and Coexistence Requirements

For each MCX flow $s \in \mathcal{S}_M$, its service requirements are summarized by:

$$\mathbf{q}_s \triangleq (\kappa_s, \Delta_s, r_s^{\min}, D_s^{\max}, J_s^{\max}, A_s^{\max}, T_s^{\text{out}}, \beta_s, \epsilon_s, \zeta_s) \quad (22)$$

where κ_s is the service class, $r_s^{\min}$ is the minimum supported payload rate, $D_s^{\max}$, $J_s^{\max}$, and $A_s^{\max}$ are the RAN-side delay, packet-delay-variation, and AoI thresholds, respectively, β_s contains the tolerated within-window violation fractions, and ϵ_s is the violation-budget vector, which are given by

$$\begin{cases} \beta_s = [\beta_s^D, \beta_s^J, \beta_s^A, \beta_s^E]^T; \\ \epsilon_s = [\epsilon_s^R, \epsilon_s^D, \epsilon_s^J, \epsilon_s^A, \epsilon_s^E]^T. \end{cases} \quad (23)$$

where β_s^q ($q \in \mathcal{Q}_s$) and ϵ_s^q ($q \in \mathcal{Q}_s$)⁺ are the tolerated within-window violation and the violation-budget for delay, jitter, AoI, and reliability, $\mathcal{Q}_s \subseteq \{D, J, A, E\}$ contains the QoS dimensions active for service s , and $\mathcal{Q}_s^+ = \{R\} \cup \mathcal{Q}_s$ denotes

the active risk-index set, where R denotes the delivered-rate violation. Setting $\beta_s^q = 0$ recovers a zero-tolerance. A positive β_s^q permits a small application-specific fraction of within-window violations.

For a successfully delivered packet, its RAN-side delay, denoted by $D_{s,n}^a$, is given as follows:

$$D_{s,n}^a = d_{s,n}^a - g_{s,n}. \quad (24)$$

Let $\mathcal{P}_s^\Delta$ denote the set of consecutive successfully delivered packet pairs in source sequence order and n^- be the preceding successfully delivered packet in source sequence order. For $(n^-, n) \in \mathcal{P}_s^\Delta$, define the packet-delay-variation magnitude, denoted by $J_{s,n}^a$, as follows:

$$J_{s,n}^a = |D_{s,n}^a - D_{s,n^-}^a|. \quad (25)$$

For a status-update flow, let

$$u_s^a(t) = \max \{ g_{s,n} : Y_{s,n}^a = 1, d_{s,n}^a \leq t \} \quad (26)$$

be the generation time of the freshest update delivered by time t . The instantaneous AoI, denoted by $AoI_s^a(t)$, is defined as follows:

$$AoI_s^a(t) = t - u_s^a(t). \quad (27)$$

The value of $u_s^a(t_s^0)$ is initialized by the freshest update delivered before the finite horizon.

Define the fully observed packet set, denoted by $\mathcal{N}_s^{\text{out}}$, as follows:

$$\mathcal{N}_s^{\text{out}} = \{ n : t_s^0 \leq g_{s,n} \leq t_s^0 + \Delta_s - T_s^{\text{out}} \}, \quad (28)$$

The arrival-time definition of $\mathcal{N}_s^{\text{out}}$ avoids outcome-dependent right censoring. Define the successful delivered subset, denoted by $\mathcal{N}_{s,a}^{\text{succ}}$, within the finite operational window as follows:

$$\mathcal{N}_{s,a}^{\text{succ}} = \{ n \in \mathcal{N}_s^{\text{out}} : Y_{s,n}^a = 1 \}. \quad (29)$$

Then, the realized delivered rate and terminal non-delivery ratio are given, respectively, as follows:

$$\begin{cases} R_{s,a}^{\Delta} &= \frac{1}{\Delta_s} \sum_{k \in \mathcal{K}_s} S_{s,a}[k]; \\ Z_{s,a}^E &= \frac{1}{|\mathcal{N}_s^{\text{out}}|} \sum_{n \in \mathcal{N}_s^{\text{out}}} (1 - Y_{s,n}^a), \end{cases} \quad (30)$$

where $\mathcal{K}_s \triangleq \{k_{l_s}, k_{l_s} + 1, \dots, k_{l_s} + K_s - 1\}$ is the mission-slot index set. The remaining mission metrics are given as follows:

$$\begin{cases} Z_{s,a}^D = \frac{1}{|\mathcal{N}_{s,a}^{\text{succ}}|} \sum_{n \in \mathcal{N}_{s,a}^{\text{succ}}} \mathbf{1} \{ D_{s,n}^a > D_s^{\max} \}; \\ Z_{s,a}^J = \frac{1}{|\mathcal{P}_s^\Delta|} \sum_{(n^-, n) \in \mathcal{P}_s^\Delta} \mathbf{1} \{ |D_{s,n}^a - D_{s,n^-}^a| > J_s^{\max} \}; \\ Z_{s,a}^A = \frac{1}{\Delta_s} \int_{t_s^0}^{t_s^0 + \Delta_s} \mathbf{1} \{ AoI_s^a(t) > A_s^{\max} \} dt. \end{cases} \quad (31)$$

If an active delay or jitter metric has no evaluable sample, the corresponding trajectory is conservatively counted as a mission violation. The finite-horizon service requirements are given as follows:

$$\begin{cases} \Pr \{ R_{s,a}^\Delta < r_s^{\min} | \mathcal{F}_{l_s}^{E2} \} \leq \epsilon_s^R; \\ \Pr \{ Z_{s,a}^q > \beta_s^q | \mathcal{F}_{l_s}^{E2} \} \leq \epsilon_s^q, \quad q \in \mathcal{Q}_s \end{cases} \quad (32)$$

where l_s is the orchestration epoch containing t_s^0 . Furthermore, we define the joint finite-horizon multi-QoS violation event, denoted by $\mathcal{V}_s^{MQ}$, as follows:

$$\mathcal{V}_s^{MQ} = \{R_{s,a}^\Delta < r_s^{\min}\} \cup \bigcup_{q \in \mathcal{Q}_s} \{Z_{s,a}^q > \beta_s^q\}. \quad (33)$$

For non-MCX flow n , define its conditional mean delivered rate, denoted by $\bar{r}_n[l]$, over Near-RT epoch l as follows:

$$\bar{r}_n[l] = \frac{1}{T_x} \mathbb{E} \left[\sum_{k \in \mathcal{K}_l} S_n[k] \middle| \mathcal{F}_l^{\text{E2}} \right] \quad (34)$$

where $S_n[k]$ is the unique payload bits for non-MCX flow n in slot k and $\mathcal{K}_l$ is the slot set associated with Near-RT epoch l . The aggregate non-MCX utility, denoted by $U_N[l]$, is given as follows:

$$U_N[l] = \sum_{n \in \mathcal{S}_N} \omega_n \log \left(1 + \frac{\bar{r}_n[l]}{r_{\text{ref}}} \right) \quad (35)$$

where $r_{\text{ref}} > 0$ is a normalization rate and $\omega_n > 0$ is a service weight. Let $U_N^0[l] > 0$ denote the reference utility under the same traffic, channel, mobility, and infrastructure realization, but with MCX-specific reservation and prioritization disabled. The coexistence loss, denoted by $\mathcal{L}_N[l]$, is given as:

$$\mathcal{L}_N[l] = \frac{[U_N^0[l] - U_N[l]]^+}{U_N^0[l]} \quad (36)$$

where $(x)^+ \triangleq \max\{0, x\}$ and the shared-network protection requirement is $\mathcal{L}_N[l] \leq \rho_{\max}$, where $\rho_{\max} \in [0, 1]$.

III. E2-CONDITIONED FINITE-HORIZON EFFECTIVE CAPACITY

A. Finite-Horizon Service Capability

Let $\bar{S}_{s,a}^{\text{pre}}[k] \geq 0$ denote the backlogged potential service available to flow s during the pre-actuation phase, and let $\bar{S}_{s,a}^{\text{post}}[k] \geq 0$ denote the backlogged potential service after candidate profile a becomes effective. For a newly admitted flow, we conservatively set $\bar{S}_{s,a}^{\text{pre}}[k] = 0$ unless a default service contract is explicitly modeled. The cumulative actuation-aware potential service, denoted by $\bar{S}_{s,a}^\Delta$, over the complete mission horizon is defined as follows:

$$\bar{S}_{s,a}^\Delta \triangleq \sum_{k=k_{l_s}}^{k_{s,a}^{\text{eff}}-1} \bar{S}_{s,a}^{\text{pre}}[k] + \sum_{k=k_{s,a}^{\text{eff}}}^{k_{l_s}+K_s-1} \bar{S}_{s,a}^{\text{post}}[k]. \quad (37)$$

Definition 1 (E2-conditioned actuation-aware FH-EC): For QoS exponent $\theta_s > 0$, the conditional service transform, denoted by $\mathcal{M}_{s,a}^{\text{FH}}(\theta_s)$, is defined as follows:

$$\mathcal{M}_{s,a}^{\text{FH}}(\theta_s) \triangleq \mathbb{E} \left[\exp \left(-\theta_s \bar{S}_{s,a}^\Delta \right) \middle| \mathcal{F}_{l_s}^{\text{E2}}, \varpi_{s,a} \right]. \quad (38)$$

The corresponding E2-conditioned finite-horizon effective capacity, denoted by $C_{s,a}^{\text{FH}}(\theta_s)$ is defined as:

$$C_{s,a}^{\text{FH}}(\theta_s) \triangleq -\frac{1}{\theta_s \Delta_s} \log \mathcal{M}_{s,a}^{\text{FH}}(\theta_s). \quad (39)$$

Note that Definition 1 presents a finite-horizon conditional log-moment measure of the service process. Unlike conventional effective capacity, it does not replace the conditional

service distribution by its long-term stationary limit [15, 16]. Instead of replacing the infinite-horizon limit by an arbitrary observation window, the proposed FH-EC preserves the operational horizon as a mission parameter and retains the initial conditional distribution induced by E2 exposure.

Definition 2: (E2-conditioned finite-horizon effective bandwidth) For $\theta_s > 0$, the finite-horizon effective bandwidth, denoted by $B_s^{\text{FH}}(\theta_s)$, is defined as follows:

$$B_s^{\text{FH}}(\theta_s) \triangleq \frac{1}{\theta_s \Delta_s} \log \mathbb{E} [\exp(\theta_s A_s(l_s, \Delta_s)) \middle| \mathcal{F}_{l_s}^{\text{E2}}] \quad (40)$$

where $A_s(l_s, \Delta_s)$ is the cumulative arrival process over the same horizon, which is given by

$$A_s(l_s, \Delta_s) = \sum_{k=k_l}^{k_l+K_s-1} A_s[k]. \quad (41)$$

Assume that, conditioned on $\mathcal{F}_l^{\text{E2}}$, the exogenous arrival process and future potential-service innovations are independent. For any backlog surplus threshold $q_s^{\text{th}} \geq 0$, the Chernoff inequality yields:

$$\Pr \left\{ A_s(l_s, \Delta_s) - \bar{S}_{s,a}^\Delta \geq q_s^{\text{th}} \middle| \mathcal{F}_l^{\text{E2}}, \varpi_{s,a} \right\} \leq \exp \left[-\theta_s q_s^{\text{th}} + \theta_s \Delta_s (B_s^{\text{FH}}(\theta_s) - C_{s,a}^{\text{FH}}(\theta_s)) \right]. \quad (42)$$

A sufficient condition for controlling the finite-window service deficit exponent is given as follows:

$$B_s^{\text{FH}}(\theta_s) \leq C_{s,a}^{\text{FH}}(\theta_s). \quad (43)$$

To guarantee a prescribed service-deficit probability, denoted by $\epsilon_s^{\text{sd}} \in (0, 1)$, a sufficient feasibility condition is given as follows:

$$B_s^{\text{FH}}(\theta_s) + \frac{\log(1/\epsilon_s^{\text{sd}})}{\theta_s \Delta_s} \leq C_{s,a}^{\text{FH}}(\theta_s) + \frac{q_s^{\text{th}}}{\Delta_s}. \quad (44)$$

Thus, Eq. (43) preserves the traditional EC interpretation, whereas Eq. (44) supplies an explicit finite-window probability margin.

To obtain a computable representation of Eq. (38), let

$$X_{s,a}[k] = (\{Z_{s,a,\ell}[k]\}_{\ell \in \mathcal{L}_a}, Z_{s,a}^{\text{cm}}[k], H_{s,a}[k]) \in \mathcal{X}_{s,a}, \quad (45)$$

denote a finite-state abstraction of the domain condition, where $Z_{s,a,\ell}[k]$ and $Z_{s,a}^{\text{cm}}[k]$ are the radio/service state over leg ℓ and transport, O-CU-UP, shared power, or common-mode failure state, respectively, $\mathcal{L}_a$ is the duplication-leg index set, and $H_{s,a}[k]$ is the HARQ/retransmission phase. The state can jointly represent quantized channel quality, interference/load level, O-DU availability, retransmission phase, and other variables governing the service process. Finite-state Markov channel models are a standard abstraction for temporally correlated wireless channels [26].

The current conditional state distribution, denoted by $\alpha_{s,a}[l]$, is given as follows:

$$\alpha_{s,a}[l] = [\Pr \{X_{s,a}[k_l] = m \middle| \mathcal{F}_l^{\text{E2}}\}]_{m=1}^{M_{s,a}}. \quad (46)$$

For phase $r \in \{\text{pre}, \text{post}\}$, we define the phase-dependent service-weighted transition kernel, denoted by $[\mathbf{G}_{s,a}^r[k](\theta_s)]_{uv}$, as follows:

$$[\mathbf{G}_{s,a}^r[k](\theta_s)]_{uv} \triangleq \mathbb{E} \left[\exp \left(-\theta_s \overline{S}_{s,a}^r[k] \right) \times \mathbf{1} \{X_{s,a}[k+1] = v\} \middle| X_{s,a}[k] = u, \mathcal{F}_{l_s}^{\text{E2}} \varpi_{s,a}^r \right], \quad (47)$$

where $\varpi_{s,a}^{\text{post}} \equiv \varpi_{s,a}$ and $\varpi_{s,a}^{\text{pre}} \equiv \varpi_{s,a}^{\text{pre}}$. For compactness, we define the following time-ordered product:

$$\prod_{k=a}^{\rightarrow b} \mathbf{G}[k] \triangleq \mathbf{G}[a] \mathbf{G}[a+1] \cdots \mathbf{G}[b]. \quad (48)$$

Define the pre-actuation kernel product, denoted by $\mathbf{H}_{s,a}^{\text{pre}}(\theta_s)$, and post-actuation kernel product, denoted by $\mathbf{H}_{s,a}^{\text{post}}(\theta_s)$, respectively, as follows:

$$\begin{cases} \mathbf{H}_{s,a}^{\text{pre}}(\theta_s) \triangleq \prod_{k=k_{l_s}}^{\rightarrow k_{s,a}^{\text{eff}}-1} \mathbf{G}_{s,a}^{\text{pre}}[k](\theta_s); \\ \mathbf{H}_{s,a}^{\text{post}}(\theta_s) \triangleq \prod_{k=k_{s,a}^{\text{eff}}}^{\rightarrow k_{l_s}+K_s-1} \mathbf{G}_{s,a}^{\text{post}}[k](\theta_s). \end{cases} \quad (49)$$

An empty product is defined as the identity matrix. Then, the exact actuation-aware FH-EC for the adopted finite-state Markov-additive model is therefore given as:

$$C_{s,a}^{\text{FH}}(\theta_s) = -\frac{1}{\theta_s \Delta_s} \log \left[\boldsymbol{\alpha}_{s,a}^T[l_s] \mathbf{H}_{s,a}^{\text{pre}}(\theta_s) \mathbf{H}_{s,a}^{\text{post}}(\theta_s) \mathbf{1} \right]. \quad (50)$$

Proposition 1 (Long-horizon consistency) Fix profile a and exponent $\theta_s > 0$. For a sequence of mission horizons $\Delta_s = K_s T_{\text{sl}}$, let d_{K_s} denote the number of pre-actuation slots and $\mathbf{H}_{K_s}^{\text{pre}}$ be the corresponding pre-actuation kernel product. Suppose that:

- 1) $\sup_{K_s} d_{K_s} \leq d_{\max} < \infty$;
- 2) After actuation, the service-weighted kernel is time homogeneous and primitive:

$$\mathbf{G}_{s,a}^{\text{post}}[k](\theta_s) = \mathbf{G}_{s,a}^{\infty}(\theta_s), \quad k \geq k_{s,a}^{\text{eff}}; \quad (51)$$

- 3) If $\mathbf{v}_{s,a}$ is the positive Perron right eigenvector of $\mathbf{G}_{s,a}^{\infty}(\theta_s)$, then there exist constants $0 < c_- \leq c_+ < \infty$ such that

$$c_- \leq \boldsymbol{\alpha}_{s,a}^T[l_s] \mathbf{H}_{K_s}^{\text{pre}}(\theta_s) \mathbf{v}_{s,a} \leq c_+, \quad \forall K_s. \quad (52)$$

Then, the long-horizon effective capacity, denoted by $C_{s,a}^{\infty}(\theta_s)$, is obtained as follows:

$$C_{s,a}^{\infty}(\theta_s) \triangleq \lim_{K_s \rightarrow \infty} C_{s,a}^{\text{FH}}(\theta_s) = -\frac{1}{\theta_s T_{\text{sl}}} \log \rho(\mathbf{G}_{s,a}^{\infty}(\theta_s)). \quad (53)$$

where $\rho(\cdot)$ denotes the spectral radius.

Proof: The proof of Proposition 1 is in Appendix A. ■

Remark 1: Proposition 1 shows that the finite E2 selection-and-actuation transient affects short mission windows but contributes only a vanishing $O(d_{s,a}^{\text{ctl}}/K_s)$ term in the long-horizon regime. Consequently, the proposed FH-EC retains the conventional effective-capacity limit without ignoring the service loss before an E2 action becomes effective.

Assume that the admitted MCX traffic satisfies the following finite-window arrival envelope:

$$A_s(l_s, \Delta_s) \leq \sigma_s + r_s \Delta_s, \quad (54)$$

where σ_s is the maximum admitted burst and r_s is the sustained payload rate. For any service-deficit threshold $q_s^{\text{th}} \geq 0$, by applying Markov's inequality, we have

$$\Pr \left\{ A_s(l_s, \Delta_s) - \overline{S}_{s,a}^{\Delta_s} \geq q_s^{\text{th}} \mid \mathcal{F}_{l_s}^{\text{E2}}, \varpi_{s,a} \right\} \leq \exp \left[-\theta_s (q_s^{\text{th}} - \sigma_s) + \theta_s \Delta_s (r_s - C_{s,a}^{\text{FH}}(\theta_s)) \right]. \quad (55)$$

Definition 3: (Finite-horizon supportable admitted rate) For a prescribed service-deficit probability ϵ_s^{sd} , candidate a supports any rate not larger than the finite-horizon supportable admitted rate, denoted by $r_{s,a}^{\text{FH}}$, which is given as follows:

$$r_{s,a}^{\text{FH}} = \left[C_{s,a}^{\text{FH}}(\theta_s) + \frac{q_s^{\text{th}} - \sigma_s}{\Delta_s} - \frac{\log(1/\epsilon_s^{\text{sd}})}{\theta_s \Delta_s} \right]^+. \quad (56)$$

B. Capability Characterization under Connectivity Diversity

The objective is not to design a new duplication mechanism, but to quantify heterogeneous connectivity options through the proposed FH-EC primitive. For a fixed E2-executable policy $\varpi_{s,a}$, we define the delivered-rate risk and the dimension-specific multi-QoS risk, respectively, as follows:

$$\begin{cases} p_{s,a}^{R,\text{FH}} \triangleq \Pr \{ R_{s,a}^{\Delta_s} < r_{s,a} \mid \mathcal{F}_{l_s}^{\text{E2}}, \varpi_{s,a} \}; \\ p_{s,a}^{q,\text{FH}} \triangleq \Pr \{ Z_{s,a}^q > \beta_q \mid \mathcal{F}_{l_s}^{\text{E2}}, \varpi_{s,a} \}. \end{cases} \quad (57)$$

where $r_{s,a} \geq r_s^{\min}$ is the finite-horizon delivered-rate target. The associated multi-QoS risk vector, denoted by $\mathbf{p}_{s,a}^{\text{FH}}$, is given by

$$\mathbf{p}_{s,a}^{\text{FH}} = \left[p_{s,a}^{R,\text{FH}}, \{ p_{s,a}^{q,\text{FH}} \}_{q \in \mathcal{Q}_s} \right]^T. \quad (58)$$

An exact Markov recursion could in principle be constructed by augmenting the state with accumulated violation counters, the number of evaluable packets or packet pairs, and the accumulated AoI violation time. However, this would produce a high-dimensional state whose size grows with the mission horizon and packet count. We therefore estimate Eq. (57) from complete finite-horizon delivery trajectories rather than claiming a low-dimensional exact recursion.

1) Lower Confidence Bound on Finite-Horizon Capability: Let $c_l = g(\mathcal{F}_{l_s}^{\text{E2}}) \in \mathcal{C}$ denote the finite E2-context label, where the context map g , the finite context set $\mathcal{C}$, and the candidate-policy library are fixed using an independent calibration dataset $\mathcal{D}^{\text{tr}}$ before $\mathcal{D}^{\text{cal}}$ is accessed. For a concurrent profile-selection vector $\bar{\mathbf{x}}$, let $e_{s,a}(\bar{\mathbf{x}}, c) \in \mathcal{E}_c^{\text{exec}}$ denote the finite execution-class label experienced by profile a of flow s in context cell c . The execution class records the active MCX service composition, aggregate resource contract, participating radio legs, fixed O-CU/O-DU policies, calibrated interference/load envelope, and common-mode failure class. The finite execution-class set $\mathcal{E}_c^{\text{exec}}$ is fixed before calibration. Define the calibration key as $j \triangleq (a, c, e, \theta_s, \mathbf{q}_s) \in \mathcal{J}$, where $\mathbf{q}_s$ contains the service class, mission horizon, and all active delivered-rate, delay, jitter, AoI, and terminal non-delivery requirements. The finite calibration library $\mathcal{J}$ is fixed before $\mathcal{D}^{\text{cal}}$ is accessed.

Let $\mathcal{W}_j$ denote a complete mission trajectory generated under calibration key j . Then, the exact cell-and-execution-class-conditioned FH-EC is given by

$$C_j^{\text{FH}} \triangleq -\frac{1}{\theta_s \Delta_s} \log \mathbb{E} \left[\exp \left(-\theta_s \bar{S}_j^{\Delta_s} \right) \mid c_{l_s} = c, e_{s,a} = e \right]. \quad (59)$$

For $q \in \mathcal{Q}_s^+$, define

$$p_j^{q,\text{FH}} \triangleq \Pr \{ \mathcal{V}_j^q \mid c_{l_s} = c, e_{s,a} = e \}, \quad (60)$$

where

$$\mathcal{V}_j^R = \{ R_{s,a}^{\Delta_s} < r_s^{\text{tgt}} \}, \quad \mathcal{V}_j^q = \{ Z_{s,a}^q > \beta_s^q \}, \quad q \in \mathcal{Q}_s. \quad (61)$$

Suppose that N_j independent complete mission trajectories are available for profile j , producing accumulated potential services $\bar{S}_j^{(1)}(\Delta_s), \dots, \bar{S}_j^{(N_j)}(\Delta_s)$. We define

$$Y_j^{(n)} = \exp \left(-\theta_s \bar{S}_j^{(n)}(\Delta_s) \right) \in (0, 1] \quad (62)$$

and

$$\hat{\mu}_j = \frac{1}{N_j} \sum_{n=1}^{N_j} Y_j^{(n)}. \quad (63)$$

For a confidence budget $\alpha_j^C \in (0, 1)$, by applying Hoeffding's inequality, we have [27]

$$\Pr_{\text{cal}} \left\{ \mathbb{E}[Y_j] \leq \hat{\mu}_j + \sqrt{\frac{\log(1/\alpha_j^C)}{2N_j}} \right\} \geq 1 - \alpha_j^C. \quad (64)$$

Because $-\log(\cdot)$ is decreasing, a valid lower confidence bound on the finite-horizon capability is

$$\underline{C}_j^{\text{FH}} = -\frac{1}{\theta_s \Delta_s} \log \left[\min \left\{ 1, \hat{\mu}_j + \sqrt{\frac{\log(1/\alpha_j^C)}{2N_j}} \right\} \right]. \quad (65)$$

Hence, we have

$$\Pr_{\text{cal}} \{ C_j^{\text{FH}} \geq \underline{C}_j^{\text{FH}} \} \geq 1 - \alpha_j^C. \quad (66)$$

2) *Statistically Calibrated Executable Profiles*: The exact finite-horizon capability and operational risks in $\phi_{s,a}[l]$ are generally unknown and must be inferred from finite data. To separate policy construction from statistical validation, the context partition, candidate-policy library, and all model hyperparameters are fixed using a training set, denoted by $\mathcal{D}^{\text{tr}}$ before the independent calibration set $\mathcal{D}^{\text{cal}}$ is accessed.

Define the simultaneous profile-validity event, denoted by $\mathcal{E}_{\text{prof}}$, as follows:

$$\mathcal{E}_{\text{prof}} \triangleq \bigcap_{j \in \mathcal{J}} \{ C_j^{\text{FH}} \geq \underline{C}_j^{\text{FH}}, \quad p_j^{q,\text{FH}} \leq \bar{p}_j^{q,\text{FH}}, \quad \forall q \in \mathcal{Q}_j^+ \}. \quad (67)$$

where $\bar{p}_j^q$ is the exact Clopper–Pearson upper confidence limit, which is given as follows:

$$\bar{p}_j^q = \begin{cases} 1, & K_{j,q} = N_j; \\ \text{Beta}^{-1}(1 - \alpha_{j,q}; K_{j,q} + 1, N_j - K_{j,q}), & K_{j,q} < N_j, \end{cases} \quad (68)$$

where $\text{Beta}^{-1}(\cdot)$ is the beta distribution, $\alpha_{j,q}$ is the one-sided confidence budget, and $K_{j,q}$ is the observed number of type- q mission-window violations.

Assigning the one-sided confidence-failure budgets, we have

$$\sum_{j \in \mathcal{J}} \left(\alpha_j^C + \sum_{q \in \mathcal{Q}_j^+} \alpha_{j,q} \right) \leq \alpha_{\text{prof}}. \quad (69)$$

By applying the individual one-sided coverage guarantees and the union bound, we obtain

$$\Pr_{\text{cal}} \{ \mathcal{E}_{\text{prof}} \} \geq 1 - \alpha_{\text{prof}}. \quad (70)$$

Because $\mathcal{E}_{\text{prof}}$ holds simultaneously over the entire finite library $\mathcal{J}$, subsequently selecting a calibrated profile after observing the current E2-context cell does not incur an additional post-selection confidence penalty. This guarantee is conditional on the calibrated context and execution class; it is not a pointwise guarantee for every raw realization of $\mathcal{F}_l^{\text{E2}}$ within the same context cell.

3) *Confidence-Calibrated E2-Executable Profiles*: For calibration key j , we define the confidence-calibrated executable profile, denoted by ϕ_j , as follows:

$$\phi_j \triangleq (j, m_a, \mathcal{I}_a, \mathbf{b}_{s,a}, \varpi_{s,a}, \underline{C}_j^{\text{FH}}, \bar{\mathbf{p}}_j^{\text{FH}}, \tau_{s,a}^{\text{ctl}}), \quad (71)$$

where

$$\bar{\mathbf{p}}_j^{\text{FH}} = [\bar{p}_j^{R,\text{FH}}, \{ \bar{p}_j^{q,\text{FH}} \}_{q \in \mathcal{Q}_s}]^T. \quad (72)$$

Let $\mathcal{A}_s^{\text{E2}}[l]$ denote the executable candidate set. At the admission epoch l_s , let c_{l_s} denote the calibrated E2-context cell and let $\mathcal{C}_{s,a}$ denote the set of context cells for which profile a has been calibrated. The capability-feasible profile set, denoted by $\Phi_s^{\text{FH}}[l]$, representing the finite candidate set, is given as follows:

$$\begin{aligned} \Phi_s^{\text{FH}}[l_s] \triangleq \{ \phi_j : a \in \mathcal{A}_s^{\text{E2}}[l_s], \quad c_{l_s} \in \mathcal{C}_{s,a}, \\ d_{s,a}^{\text{ctl}} < K_s, \quad r_{s,a}^{\text{FH}} \geq r_s^{\text{min}}, \\ B_s^{\text{FH}}(\theta_s) \leq \underline{C}_{s,a}^{\text{FH}}(\theta_s), \\ B_s^{\text{FH}}(\theta_s) + \frac{\log(1/\epsilon_s^{\text{sd}})}{\theta_s \Delta_s} \leq \underline{C}_{s,a}^{\text{FH}}(\theta_s) + \frac{q_s^{\text{th}}}{\Delta_s}, \\ \bar{p}_{s,a}^{q,\text{FH}} \leq \epsilon_s^q, \quad q \in \mathcal{Q}_s^+, \\ \tau_{s,a}^{\text{ctl}} \leq \min\{T_x, \Delta_s\} \}. \end{aligned} \quad (73)$$

IV. FH-EC-DRIVEN MCX PROFILE ORCHESTRATION IN O-RAN

A. Joint Profile Selection Under Shared Resources

At orchestration epoch l , partition the active MCX flows into the following sets:

$$\begin{cases} \mathcal{S}_M[l] = \mathcal{S}_M^{\text{lock}}[l] \cup \mathcal{S}_M^{\text{new}}[l]; \\ \mathcal{S}_M^{\text{lock}}[l] \cap \mathcal{S}_M^{\text{new}}[l] = \emptyset, \end{cases} \quad (74)$$

where $\mathcal{S}_M^{\text{lock}}[l]$ contains previously admitted flows whose mission windows have not expired, and $\mathcal{S}_M^{\text{new}}[l]$ contains flows requesting admission at epoch l . For every $s \in \mathcal{S}_M^{\text{lock}}[l]$, let a_s^{lock} denote its profile selected at admission. The corresponding profile-selection variable is not re-optimized during $\mathcal{T}_s$.

The aggregate resource contract, denoted by $\mathbf{b}^{\text{lock}}[l]$, of the locked flows is given as follows:

$$\mathbf{b}^{\text{lock}}[l] \triangleq \sum_{s \in \mathcal{S}_M^{\text{lock}}[l]} \mathbf{b}_{s, a_s^{\text{lock}}}. \quad (75)$$

For each new flow $s \in \mathcal{S}_M^{\text{new}}[l]$, define the eligible nonzero candidate set, denoted by $\mathcal{A}_s^+[l]$, as follows:

$$\mathcal{A}_s^+[l] \triangleq \{a : \phi_j \in \Phi_s^{\text{FH}}[l]\}. \quad (76)$$

A zero profile $a = 0$ is added only for a new flow, with $\mathbf{b}_{s,0} = \mathbf{0}$ and $V_{s,0} = 0$. Hence, we have $\mathcal{A}_s[l] = \mathcal{A}_s^+[l] \cup \{0\}$, $s \in \mathcal{S}_M^{\text{new}}[l]$. The zero profile denotes admission rejection and cannot be used to withdraw a previously certified mission-horizon contract.

Let c_l denote the calibrated E2-context cell at epoch l . The context descriptor includes the active MCX service composition used to construct the profile library. Let

$$\mathbf{B}^{\text{av}}[l] = [B_1^{\text{av}}[l], \dots, B_R^{\text{av}}[l]]^T \in \mathbb{R}_+^R \quad (77)$$

denote the resource vector available at epoch l , after accounting for O-DU degradation and resources reserved outside the considered orchestration problem. Define $\mathcal{G}_c$ as the finite set of aggregate MCX resource contracts attainable in context cell c . We have $\mathbf{0} \in \mathcal{G}_c$. Fix a non-MCX allocation rule ψ_N that is used during both calibration and runtime execution. For an aggregate MCX contract $\mathbf{b} \in \mathcal{G}_c$, let $\bar{r}_{n,c}^{\psi_N}(\mathbf{b})$ denote the true conditional mean rate of non-MCX flow n when $\mathbf{b}$ is enforced in context cell c and ψ_N allocates the residual resources. We define the corresponding non-MCX utility, denoted by $U_{N,c}^{\psi_N}(\mathbf{b})$, as follows:

$$U_{N,c}^{\psi_N}(\mathbf{b}) \triangleq \sum_{n \in \mathcal{S}_N} \omega_n \log \left(1 + \frac{\bar{r}_{n,c}^{\psi_N}(\mathbf{b})}{r_{\text{ref}}} \right). \quad (78)$$

Let $U_{N,c}^0$ denote the context-matched reference utility with MCX-specific reservation and prioritization disabled. The policy-induced coexistence-supportable set, denoted by $\mathcal{B}_{N,c}^{\psi_N}$, is given by

$$\mathcal{B}_{N,c}^{\psi_N} \triangleq \left\{ \mathbf{b} \in \mathcal{G}_c : U_{N,c}^{\psi_N}(\mathbf{b}) \geq (1 - \rho_{\max}) U_{N,c}^0, \right. \\ \left. \bar{r}_{n,c}^{\psi_N}(\mathbf{b}) \geq \underline{r}_n^N, \quad n \in \mathcal{S}_N^{\text{prot}} \right\}. \quad (79)$$

Since ψ_N provides an explicit feasible non-MCX allocation, we have

$$\mathcal{B}_{N,c}^{\psi_N} \subseteq \mathcal{B}_{N,c}^{\text{coex}} \cap \mathcal{G}_c \quad (80)$$

where $\mathcal{B}_{N,c}^{\text{coex}}$ is the exact set that satisfies the non-MCX requirements given the existence of a feasible residual policy. For each context cell c and aggregate MCX contract $\mathbf{b} \in \mathcal{G}_c$, let $\{R_{n,c}^{(j)}(\mathbf{b})\}_{j=1}^{N_{c,b}^0}$ be independent calibration realizations of the epoch-level delivered rate of non-MCX flow n under the fixed residual-resource policy ψ_N . Assume that

$$0 \leq R_{n,c}^{(j)}(\mathbf{b}) \leq r_n^{\max}, \quad (81)$$

where $r_n^{\max}$ is a known physical upper bound. The sample mean, denoted by $\hat{r}_{n,c}(\mathbf{b})$, is derived as follows:

$$\hat{r}_{n,c}(\mathbf{b}) = \frac{1}{N_{c,b}} \sum_{j=1}^{N_{c,b}} R_{n,c}^{(j)}(\mathbf{b}). \quad (82)$$

For a one-sided confidence budget $\alpha_{n,c,b}^-$, Hoeffding's inequality gives the following lower confidence bound (LCB):

$$\underline{r}_{n,c}(\mathbf{b}) = \left[\hat{r}_{n,c}(\mathbf{b}) - r_n^{\max} \sqrt{\frac{\log(1/\alpha_{n,c,b}^-)}{2N_{c,b}}} \right]^+. \quad (83)$$

Likewise, let $\{R_{n,c}^{0,(j)}\}_{j=1}^{N_c^0}$ denote the context-matched reference-rate realizations with MCX-specific reservation and prioritization disabled. We define

$$\hat{r}_{n,c}^0 = \frac{1}{N_c^0} \sum_{j=1}^{N_c^0} R_{n,c}^{0,(j)} \quad (84)$$

and the upper confidence bound (UCB) as follows:

$$\bar{r}_{n,c}^0 = \min \left\{ r_n^{\max}, \hat{r}_{n,c}^0 + r_n^{\max} \sqrt{\frac{\log(1/\alpha_{n,c}^0)}{2N_c^0}} \right\}. \quad (85)$$

Since the utility function is coordinate-wise increasing, define

$$\begin{cases} \underline{U}_{N,c}(\mathbf{b}) &= \sum_{n \in \mathcal{S}_N} \omega_n \log \left(1 + \frac{\underline{r}_{n,c}(\mathbf{b})}{r_{\text{ref}}} \right); \\ \bar{U}_{N,c}^0 &= \sum_{n \in \mathcal{S}_N} \omega_n \log \left(1 + \frac{\bar{r}_{n,c}^0}{r_{\text{ref}}} \right). \end{cases} \quad (86)$$

Assign the confidence budgets such that

$$\sum_c \left[\sum_{n \in \mathcal{S}_N} \alpha_{n,c}^0 + \sum_{\mathbf{b} \in \mathcal{G}_c} \sum_{n \in \mathcal{S}_N} \alpha_{n,c,b}^- \right] \leq \alpha_N. \quad (87)$$

We define the simultaneous non-MCX validity event, denoted by $\mathcal{E}_N$, as follows:

$$\mathcal{E}_N \triangleq \bigcap_c \left[\bigcap_{\mathbf{b} \in \mathcal{G}_c} \bigcap_{n \in \mathcal{S}_N} \{r_{n,c}^{\psi_N}(\mathbf{b}) \geq \underline{r}_{n,c}(\mathbf{b})\} \right. \\ \left. \cap \bigcap_{n \in \mathcal{S}_N} \{r_{n,c}^0 \leq \bar{r}_{n,c}^0\} \right]. \quad (88)$$

By Hoeffding's inequality and the union bound, we obtain

$$\Pr_{\text{cal}}\{\mathcal{E}_N\} \geq 1 - \alpha_N. \quad (89)$$

Thus, the finite-sample calibration-safe set, denoted by $\hat{\mathcal{G}}_c^{\text{safe}}$, is derived as follows:

$$\hat{\mathcal{G}}_c^{\text{safe}} \triangleq \left\{ \mathbf{b} \in \mathcal{G}_c : \underline{U}_{N,c}(\mathbf{b}) \geq (1 - \rho^{\max}) \bar{U}_{N,c}^0, \right. \\ \left. \underline{r}_{n,c}(\mathbf{b}) \geq r_n^N, \quad n \in \mathcal{S}_N^{\text{prot}} \right\}. \quad (90)$$

The origin-anchored calibration-safe MCX budget, denoted by $\mathbf{b}_c^M$, is selected as follows:

$$\mathbf{b}_c^M \in \arg \max_{\mathbf{b} \in \mathcal{G}_c} \mathbf{w}^T \mathbf{b} \quad (91) \\ \text{s.t.} \quad \{\mathbf{b} \in \mathcal{G}_c : \mathbf{0} \preceq \mathbf{b} \preceq \mathbf{b}\} \subseteq \hat{\mathcal{G}}_c^{\text{safe}}$$

where $w \succeq 0$ is the resource-box weight. If no nonzero budget satisfies Eq. (91), the zero budget is used. On $\mathcal{E}_N$, we have

$$\{\mathbf{b} \in \mathcal{G}_c : \mathbf{0} \preceq \mathbf{b} \preceq \mathbf{b}_c^M\} \subseteq \mathcal{B}_{N,c}^{\psi_N} \subseteq \mathcal{B}_{N,c}^{\text{coex}}. \quad (92)$$

Hence, the runtime constraint $\mathbf{b}^M(\mathbf{x}) \preceq \bar{\mathbf{b}}_{c_l}^M$ is a calibration-supported inner restriction of the exact coexistence region over the finite set of attainable profile contracts.

B. Joint Profile-Selection Problem and Finite-Library Profile Selection

For every confidence-screened nonzero profile, we define the conservative FH-EC surplus as follows:

$$\Delta \underline{C}_j^{\text{FH}} \triangleq \underline{C}_j^{\text{FH}}(\theta_s) - B_s^{\text{FH}}(\theta_s) \geq 0. \quad (93)$$

The secondary profile value, denoted by $V_{s,j}$, is given by

$$V_{s,j} = \omega_s \Delta \underline{C}_j^{\text{FH}} - \lambda_b \left\| \tilde{\mathbf{b}}_{s,a} \right\|_1 - \lambda_\tau \tau_{s,a}^{\text{ctl}}, \quad (94)$$

where $\tilde{\mathbf{b}}_{s,a}$ is the component-wise normalized resource vector. Thus, λ_b penalizes resource consumption rather than profile reconfiguration. For every new flow $s \in \mathcal{S}_M^{\text{new}}[l]$ and $a \in \mathcal{A}_s[l]$, let $x_{s,j}[l] \in \{0, 1\}$ indicate profile selection. For every candidate profile, define

$$\bar{x}_{s,j}[l] \triangleq \begin{cases} 1, & s \in \mathcal{S}_M^{\text{lock}}[l], \\ x_{s,j}[l], & s \in \mathcal{S}_M^{\text{new}}[l], \\ 0, & \text{otherwise.} \end{cases} \quad (95)$$

Define the total MCX contract after admitting the new flows as:

$$\mathbf{b}^M(\mathbf{x}; l) = \sum_s \sum_{j \in \mathcal{J}} \mathbf{b}_j \bar{x}_{s,j}[l]. \quad (96)$$

To prioritize MCX admission over secondary resource efficiency, we define

$$\tilde{V}_{s,j} \triangleq \Gamma \omega_s^{\text{adm}} \mathbf{1}\{a \neq 0\} + V_{s,j}, \quad (97)$$

where $\Gamma > 0$ is selected sufficiently large so that admission priority dominates the secondary profile value and ω_s^{adm} is the admission priority. Equivalently, Eq. (97) may be implemented as a lexicographic two-stage objective. Let $\mathcal{H}_l$ denote the precomputed set of minimal forbidden profile combinations at epoch l . A profile combination belongs to $\mathcal{H}_l$ if its induced execution class is absent from the calibrated library $\mathcal{J}$, or if it violates a stored resource, interference, load, policy, or common-mode-failure envelope.

The FH-EC profile orchestration problem is formulated as follows:

$$\mathbf{P1}[l] : \max_{\{x_{s,j}[l]\}} \sum_{s \in \mathcal{S}_M^{\text{new}}[l]} \sum_{j \in \mathcal{J}} \tilde{V}_{s,j} x_{s,j}[l] \quad (98a)$$

$$\text{s.t.} \quad \sum_{a \in \mathcal{A}_s[l]} x_{s,j}[l] = 1, \quad s \in \mathcal{S}_M^{\text{new}}[l], \quad (98b)$$

$$\mathbf{b}^M(\mathbf{x}; l) \preceq \underline{\mathbf{B}}^{\text{av}}[l], \quad (98c)$$

$$\mathbf{b}^M(\mathbf{x}; l) \preceq \mathbf{B}_{c_l}^M, \quad (98d)$$

$$\sum_{(s,j) \in H} \bar{x}_{s,j}[l] \leq |H| - 1, \quad H \in \mathcal{H}_l, \quad (98e)$$

$$x_{s,j}[l] \in \{0, 1\}. \quad (98f)$$

where $\underline{\mathbf{B}}^{\text{av}}[l]$ is the component-wise guaranteed lower envelope over the next finite operational window.

Problem **P1**[l] is a finite 0-1 linear program because all candidate-dependent capability, risk, delay, and resource terms are precomputed constants. Constraint Eq. (98c) enforces instantaneous physical-resource availability, whereas Eq. (98d) restricts the total MCX contract to the calibration-safe inner region for non-MCX services. Previously admitted mission-horizon contracts remain fixed and appear through $\mathbf{b}^{\text{lock}}[l]$. Algorithm 1 summarizes the actuation-aware FH-EC MCX profile orchestration in O-RAN. The proposed orchestration separates capability generation from real-time decision making. FH-EC estimation and certificate construction are performed offline, while the online Near-RT RIC only solves a finite-dimensional profile selection problem. Therefore, the real-time complexity is determined by the number of executable profiles rather than the instantaneous radio state space.

C. Simultaneous Post-Selection Validity of Finite-Library O-RAN Service Contracts

Definition 4 (Calibration-compatible O-RAN execution): For a feasible profile-selection vector $\mathbf{x}$ returned by **P1**[l], let

$$j_s(\mathbf{x}, l) \triangleq (a_s, c_l, e_{s,a_s}(\bar{\mathbf{x}}, c_l), \theta_s, \mathbf{q}_s) \quad (99)$$

denote the runtime calibration key of selected nonzero profile a_s . The execution is calibration compatible if:

- 1) $j_s(\mathbf{x}, l) \in \mathcal{J}$ for every selected nonzero profile;
- 2) The stored O-CU/O-DU policy, participating radio legs, resource contract, and profile identifier are preserved over the complete mission window, and the realized observation age, actuation delay, load, interference, and infrastructure state remain within the corresponding calibrated execution-class envelope;

3) Conditioned on calibration key j , the complete calibration trajectories $\{W_j^{(n)}\}_{n=1}^{N_j}$ and an independent future mission trajectory W_j^{run} are identically distributed, and the calibration trajectories are mutually independent.

If any runtime invariant in condition 2) is violated, the corresponding statistical certificate is declared invalid and the profile is rejected or transferred to a separately calibrated fallback class.

Theorem 1 (Simultaneous post-selection validity of finite-library O-RAN service contracts): Consider any post-calibration orchestration epoch l . Suppose that:

- 1) The E2-context map, finite execution-class partition, candidate-policy library, service requirements, profile screening rules, residual non-MCX policy ψ_N , and all hyperparameters are fixed before $\mathcal{D}^{\text{cal}}$ is accessed;
- 2) The calibration-runtime exchangeability condition in Definition 4 holds for every calibration key $j \in \mathcal{J}$;
- 3) The confidence-failure budgets satisfy:

$$\sum_{j \in \mathcal{J}} \left(\alpha_j^C + \sum_{q \in \mathcal{Q}_j^+} \alpha_{j,q} \right) \leq \alpha_{\text{prof}}, \quad (100)$$

Algorithm 1 Actuation-Aware FH-EC MCX Profile Orchestration

- 1: **Offline profile construction:**
 - 2: Fix the E2-context partition, finite execution-class map, and E2-executable policy library.
 - 3: Construct pre- and post-actuation service kernels.
 - 4: Compute the actuation-aware FH-EC in Eq. (50).
 - 5: Obtain simultaneous capability LCBs and operational-risk UCBs.
 - 6: Construct $\mathbf{b}_c^M$, the calibrated execution-class library, and the forbidden-hyperedge sets $\{\mathcal{H}_l\}$.
 - 7: **Online admission and orchestration at epoch l :**
 - 8: Receive the latest E2 information $\mathcal{F}_l^{\text{E2}}$.
 - 9: Identify the calibrated context cell c_l .
 - 10: Retain the locked profiles of $\mathcal{S}_M^{\text{lock}}[l]$.
 - 11: Retrieve, rather than re-estimate, the confidence-screened profile sets for $\mathcal{S}_M^{\text{new}}[l]$.
 - 12: Solve $\mathbf{P1}[l]$ in Eq. (98).
 - 13: Issue the E2 control actions associated with the selected profiles.
 - 14: Verify the induced execution-class label and monitor the stored policy, resource, load, interference, and actuation-delay envelopes throughout the mission window.
 - 15: After actuation, lock each admitted profile until its mission window expires.
 - 16: Release its reserved resources at the end of the mission window.
-

and the non-MCX budgets satisfy the simultaneous condition;

4) $\mathbf{P1}[l]$ enforces the physical-resource constraint, the calibration-safe non-MCX budget, profile locking, and the forbidden-hyperedge constraints, and returns only calibration-compatible profile combinations.

With calibration confidence at least $(1 - \alpha_{\text{prof}} - \alpha_N)$, every selected nonzero profile a_s simultaneously satisfies:

$$\begin{cases} C_{j_s(\mathbf{x}, l)}^{\text{FH}} & \geq \underline{C}_{j_s(\mathbf{x}, l)}^{\text{FH}}; \\ p_{j_s(\mathbf{x}, l)}^{q, \text{FH}} & \leq \bar{p}_{j_s(\mathbf{x}, l)}^{q, \text{FH}} \leq \epsilon_s^q, \quad q \in \mathcal{Q}_s^+; \\ \mathbf{b}^M(\mathbf{x}; l) & \leq \mathbf{b}^{\text{av}}[l]; \\ U_{N, c_l}^{\psi_N}(\mathbf{b}^M(\mathbf{x}; l)) & \geq (1 - \rho^{\max})U_{N, c_l}^0; \\ r_{n, c_l}^{\psi_N}(\mathbf{b}^M(\mathbf{x}; l)) & \geq r_n^N, \quad n \in \mathcal{S}_N^{\text{prot}}. \end{cases} \quad (101)$$

The conditional joint multi-QoS mission risk of each selected MCX flow is bounded as follows:

$$\Pr \{ \mathcal{V}_s^{\text{MQ}} \mid c_{l_s} = c_l, e_{s, a_s} = e_{s, a_s}(\bar{\mathbf{x}}, c_l) \} \leq \epsilon_s^R + \sum_{q \in \mathcal{Q}_s} \epsilon_s^q. \quad (102)$$

A prescribed joint mission-risk target ϵ_s^{MQ} is satisfied whenever the following inequality holds:

$$\epsilon_s^R + \sum_{q \in \mathcal{Q}_s} \epsilon_s^q \leq \epsilon_s^{\text{MQ}}. \quad (103)$$

Proof: The proof of Theorem 1 is in Appendix B. ■

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Logical E2 domains	4
O-DUs per domain	3
Slot duration T_{sl}	1 ms
Mission horizon Δ_s	0.1–5 s
QoS exponent θ	10^{-3} – 10^{-1}
PHY BLER	10^{-5}
PDCP duplication legs	2
E2 observation age	1–500 ms
Actuation-delay bound/range	1–20 slots
SC/DP legs	1/2

V. PERFORMANCE EVALUATIONS

A. Simulation Setup and Baselines

We evaluate the proposed FH-EC-driven MCX orchestration framework in a shared O-RAN scenario. The network simultaneously supports mission-critical MCX services and ordinary mobile services over shared radio and transport resources. The wireless service process is modeled as a finite-state Markov-additive process. The service state captures both temporal channel variations and correlated connectivity failures among duplicated transmission legs. Unless otherwise specified, the major simulation parameters are listed in Table I.

The finite-horizon capability calculation, confidence-calibrated profile library, non-MCX-safe budget, and mixed-integer profile selector are implemented in MATLAB. Packet generation, queueing, link adaptation, HARQ, infrastructure impairment, and end-to-end packet delivery are implemented in ns-3 with 5G-LENA [28–30]. 5G-LENA provides an end-to-end NR system-level simulation framework and has been calibrated against 3GPP reference scenarios.

Three E2 exposure configurations are considered. The rich configuration exposes per-flow or fine-grained load, channel-quality, queue, delivery, and node-availability information together with the broadest simulated action set. The moderate configuration exposes aggregated measurements and a reduced set of resource and association actions. The sparse configuration exposes only cell-level measurements and coarse association control.

The following schemes use the same radio resources and packet-level simulator.

- **Static-QPP:** A fixed MCX reservation and strict-priority scheduler, with its reservation tuned on the training set.
- **LT-EC:** The same finite candidate policies and test-time selector, but the finite-horizon capability coordinate is replaced by a stationary long-term EC estimate.
- **FH-SC:** The proposed finite-horizon profile construction and orchestration with DP disabled.
- **SNC-Delay:** An ORANUS-inspired delay-tail provisioning scheme using the same radio deployment and traffic traces, but without explicit jitter, AoI, and terminal-error constraints [19].
- **Proposed E2-FH:** The complete E2-conditioned finite-profile framework with adaptive SC/DP selection and the calibration-safe non-MCX resource budget.

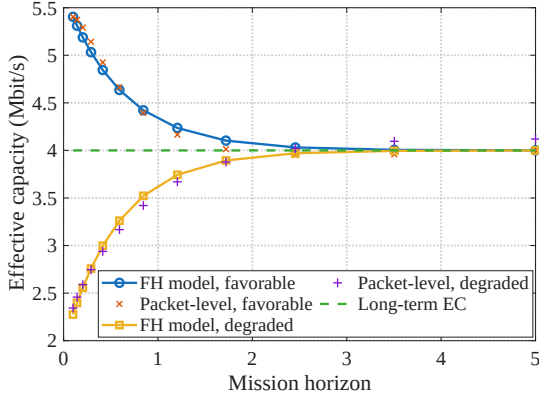


Fig. 2. Finite-horizon capability vs. mission horizon Δ_s (s).

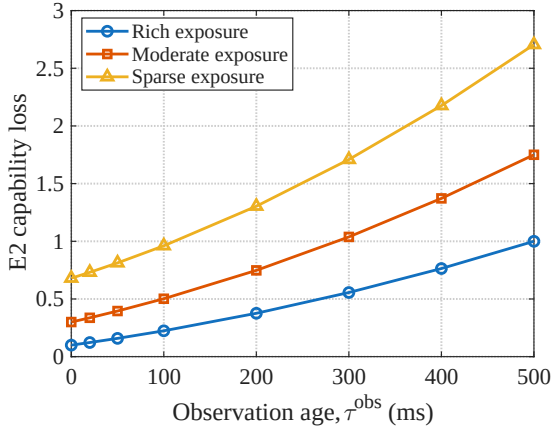


Fig. 3. E2 capability loss vs. observation age.

- **Full-State Oracle:** A reference controller with current full-state information, zero observation and actuation delay, and access to complete simulated candidate set.

B. Results and Discussion

For each mission horizon Δ_s , independent service trajectories are generated according to the Markov-additive service model. Figure 2 compares the analytical finite-horizon capability in Eq. (39), with the direct packet-level estimation and conventional long-term effective capacity as functions of the mission horizon Δ_s under favorable and degraded initial E2-conditioned states. The stationary long-term EC is included as the reference. The model follows the packet-level exponential service transform over the tested horizons. We can observe from Fig. 2 that at short horizon $\Delta_s=1$, the favorable and degraded curves differ by about 0.88 Mbit/s, confirming that the initial E2-conditioned state cannot be replaced by a stationary average. As shown in Fig. 2, as the horizon increases, the relative gap between finite-horizon and conventional EC decreases, providing numerical support for Proposition 1. Fig. 2 also shows that FH-EC captures the transient service capability within short mission windows, while gradually converging to the conventional EC when the horizon becomes sufficiently long.

O-RAN introduces additional uncertainty through E2 observation latency and control actuation delay. Fig. 3 evaluates

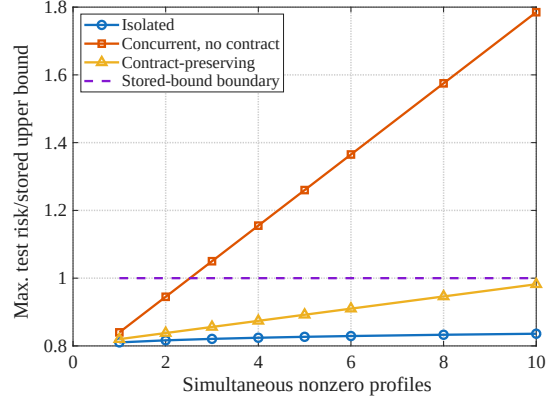


Fig. 4. Largest one-sided test upper confidence limit vs. stored profile upper bound.

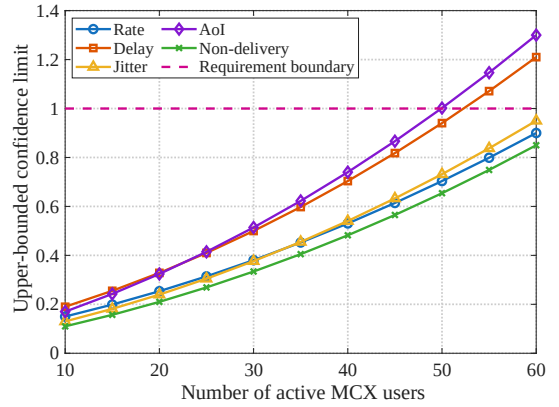


Fig. 5. Normalized one-sided risk upper confidence bound vs. number of active MCX users.

the impact of E2 exposure on the finite-horizon capability as the observation age τ_{obs} varies. We can observe from Fig. 3 that the E2 capability loss increases as the observation age increases. As shown in Fig. 3, rich exposure produces the smallest loss because it retains the most informative observation map and the largest executable action set. We can observe from Fig. 3 that outdated E2 information reduces the available MCX capability because the selected profile may no longer match the current network state.

Figure 4 plots the largest one-sided test upper confidence limit among its active operational risks over the corresponding stored profile upper bound to verify the contract-preserving composability assumption. This ratio implies a bound-validity diagnostic. The isolated execution first checks each profile individually. Concurrent execution without enforcing the characterized contracts exceeds the stored bound when more profiles are active. Under contract-preserving execution, the ratio remains no larger than one. Any profile combination that still exceeds its stored bound is excluded by an incompatibility constraint or recalibrated as a joint profile tuple.

Figure 5 plots the normalized one-sided risk upper confidence limit of each active delivered-rate, delay, jitter, AoI, and terminal non-delivery risk divided by its requirement with varying numbers of active MCX users. A value no larger than one means that the corresponding pre-existing requirement

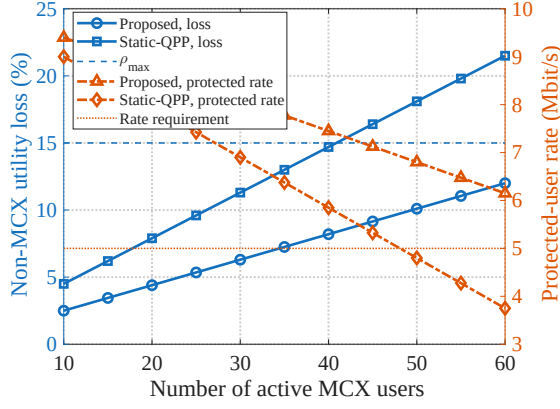


Fig. 6. Non-MCX utility loss and protected-user rate vs. number of active MCX users.

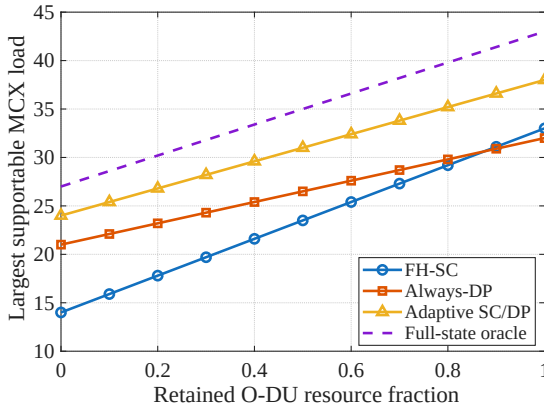


Fig. 7. Maximum admissible MCX load vs. the retained resource fraction of the degraded O-DU.

is satisfied. We can observe from Fig. 5 that the proposed method satisfies all active dimensions up to 50 simultaneous MCX users. It verifies that the improvement is not obtained by reducing delay while silently violating AoI, jitter, or terminal non-delivery requirements.

Figure 6 presents the corresponding non-MCX utility loss and the delivered rate of protected ordinary users. At the proposed method's largest supportable MCX load, the utility loss is below $\rho_{\max}$. Static-QPP either violates the protection conditions at high MCX load or reserves excess resources at low load, whereas the proposed method adapts the selected contracts within $\bar{b}_{cl}^M$. This validates the safe-budget construction in Eq. (91) and Theorem 1. In addition, Fig. 7 plots the largest supportable admissible MCX load as one O-DU is progressively degraded while comparing with SC-only, always-DP, and the adaptive SC/DP selection. We can observe from Fig. 7 that always-DP improves delivery diversity under severe independent degradation but admits fewer flows under normal operation because every copy consumes radio resources.

VI. CONCLUSIONS

We developed an E2-conditioned finite-horizon effective capacity orchestration framework that provides a unified capability abstraction for executable O-RAN configurations. Particularly, we first derived a correlation-aware Markov-additive

service model to characterize heterogeneous connectivity options. We then proposed the FH-EC expression under finite-blocklength transmission. The resulting FH-EC capability was transformed into confidence-calibrated service profiles, enabling trustworthy profile selection and realization through the Near-RT RIC. Based on the FH-EC capability abstraction, we developed an orchestration framework incorporating contract certification, shared-resource protection, and capability-aware profile selection. Coupled MATLAB/ns-3 evaluations supported the analytical finite-horizon characterization and demonstrated the effects of E2 observation age, actuation delay, concurrent profile execution, and O-DU degradation.

APPENDIX A PROOF OF PROPOSITION 1

For notational simplicity, we define

$$\begin{cases} d & \triangleq d_{s,a}^{\text{ctl}}, \\ \mathbf{H} & \triangleq \mathbf{H}_{s,a}^{\text{pre}}(\theta_s); \\ \mathbf{G} & \triangleq \mathbf{G}_{s,a}^{\infty}(\theta_s); \\ \boldsymbol{\alpha} & \triangleq \boldsymbol{\alpha}_{s,a}[l_s]. \end{cases} \quad (104)$$

Using Eq. (50), we have

$$C_{s,a}^{\text{FH}}(\theta_s) = -\frac{1}{\theta_s K_s T_{\text{sl}}} \log [\boldsymbol{\alpha}^T \mathbf{H} \mathbf{G}^{K_s-d} \mathbf{1}]. \quad (105)$$

Since $\mathbf{G}$ is primitive, the Perron–Frobenius theorem guarantees a positive spectral radius ρ , positive right and left Perron eigenvectors $\mathbf{v}$ and $\mathbf{u}$, and the normalization $\mathbf{u}^T \mathbf{v} = 1$. Moreover, we have

$$\mathbf{G}^n = \rho^n \mathbf{v} \mathbf{u}^T + o(\rho^n), \quad n \rightarrow \infty. \quad (106)$$

Therefore, we can obtain

$$\boldsymbol{\alpha}^T \mathbf{H} \mathbf{G}^{K_s-d} \mathbf{1} = \rho^{K_s-d} (\boldsymbol{\alpha}^T \mathbf{H} \mathbf{v}) (\mathbf{u}^T \mathbf{1}) [1 + o(1)]. \quad (107)$$

The coefficient in Eq. (107) is strictly positive because of $\mathbf{u}^T \mathbf{1} > 0$. By taking the logarithm, we have

$$\begin{aligned} \log [\boldsymbol{\alpha}^T \mathbf{H} \mathbf{G}^{K_s-d} \mathbf{1}] \\ = (K_s - d) \log \rho + \log [(\boldsymbol{\alpha}^T \mathbf{H} \mathbf{v}) (\mathbf{u}^T \mathbf{1})] + o(1). \end{aligned} \quad (108)$$

Substituting Eq. (108) into Eq. (105), using $d/K_s \rightarrow 0$, and letting $K_s \rightarrow \infty$, we obtain

$$\lim_{K_s \rightarrow \infty} C_{s,a}^{\text{FH}}(\theta_s) = -\frac{1}{\theta_s T_{\text{sl}}} \log \rho, \quad (109)$$

which completes the proof of Proposition 1.

APPENDIX B PROOF OF THEOREM 1

Define the simultaneous calibration event $\mathcal{E} \triangleq \mathcal{E}_{\text{prof}} \cap \mathcal{E}_N$. From the simultaneous profile-calibration result, we have $\Pr_{\text{cal}}\{\mathcal{E}_{\text{prof}}\} \geq 1 - \alpha_{\text{prof}}$, and $\Pr_{\text{cal}}\{\mathcal{E}_N\} \geq 1 - \alpha_N$. By the union bound, we obtain

$$\Pr_{\text{cal}}\{\mathcal{E}\} \geq 1 - \alpha_{\text{prof}} - \alpha_N. \quad (110)$$

On $\mathcal{E}_{\text{prof}}$, the capability lower bound and all operational-risk upper bounds hold simultaneously for every calibration

key $j \in \mathcal{J}$. By Definition 4 and the forbidden-hyperedge constraints, every nonzero profile selected by $\mathbf{P1}[l]$ induces a calibration key $j_s(\mathbf{x}, l) \in \mathcal{J}$. Consequently, we have

$$C_{j_s(\mathbf{x}, l)}^{\text{FH}} \geq \underline{C}_{j_s(\mathbf{x}, l)}^{\text{FH}}, \quad (111)$$

and

$$p_{j_s(\mathbf{x}, l)}^{q, \text{FH}} \leq \bar{p}_{j_s(\mathbf{x}, l)}^{q, \text{FH}}, \quad q \in \mathcal{Q}_s^+. \quad (112)$$

Because these inequalities hold simultaneously over the entire finite library, selecting the calibration key after observing the current E2-context cell and solving $\mathbf{P1}[l]$ does not introduce an additional post-selection confidence penalty. The profile-screening conditions then imply

$$\bar{p}_{j_s(\mathbf{x}, l)}^{q, \text{FH}} \leq \epsilon_s^q, \quad q \in \mathcal{Q}_s^+. \quad (113)$$

Constraint in Eq. (98c) directly proves $\mathbf{b}^M(\mathbf{x}; l) \preceq \mathbf{b}^{\text{av}}[l]$. On $\mathcal{E}_N$, the simultaneous non-MCX rate bounds, the definition of the calibration-safe set, and constraint Eq. (98d) give

$$U_{N, c_l}^{\psi_N}(\mathbf{b}^M(\mathbf{x}; l)) \geq (1 - \rho^{\max}) U_{N, c_l}^0, \quad (114)$$

and

$$r_{n, c_l}^{\psi_N}(\mathbf{b}^M(\mathbf{x}; l)) \geq r_n^{N, \text{prot}}, \quad n \in \mathcal{S}_N^{\text{prot}}. \quad (115)$$

Finally, by applying the union bound to the delivered-rate and active dimension-specific operational violation events, we have

$$\begin{aligned} \Pr_{\text{run}} \{ \mathcal{V}_s^{\text{MQ}} \mid c_{l_s} = c_l, e_{s, a_s} = e_{s, a_s}(\bar{\mathbf{x}}, c_l) \} \\ \leq p_{j_s(\mathbf{x}, l)}^{R, \text{FH}} + \sum_{q \in \mathcal{Q}_s} p_{j_s(\mathbf{x}, l)}^{q, \text{FH}} \leq \epsilon_s^R + \sum_{q \in \mathcal{Q}_s} \epsilon_s^q. \end{aligned} \quad (116)$$

Combining this result with Eq. (110) proves Theorem 1.

REFERENCES

- [1] B. Agarwal, R. Irmer, D. Lister, and G.-M. Muntean, "Open RAN for 6G networks: Architecture, use cases and open issues," *IEEE Communications Surveys & Tutorials*, vol. 28, pp. 2881–2924, 2026.
- [2] B. Brik, H. Chergui, L. Zanzi, F. Devoti, A. Ksentini, M. S. Siddiqui, X. Costa-Pérez, and C. Verikoukis, "Explainable AI in 6G O-RAN: A tutorial and survey on architecture, use cases, challenges, and future research," *IEEE Communications Surveys & Tutorials*, vol. 27, no. 5, pp. 2826–2859, 2025.
- [3] 3rd Generation Partnership Project, "Mission critical services common requirements (MCCoRe); stage 1," 3GPP TS 22.280, ver. 19.7.0, Release 19, Jan. 2026.
- [4] B. Karaman, I. Basturk, F. Kara, E. Zeydan, E. A. Beyazit, S. Taskin, E. Björnson, and H. Yanikomeroglu, "On-demand HAPS-assisted communication system for public safety in emergency and disaster response," *IEEE Communications Magazine*, vol. 63, no. 12, pp. 54–61, 2025.
- [5] Y. Polyanskiy, H. V. Poor, and S. Verdú, "Channel coding rate in the finite blocklength regime," *IEEE Transactions on Information Theory*, vol. 56, no. 5, pp. 2307–2359, 2010.
- [6] R. D. Yates, Y. Sun, D. R. Brown, S. K. Kaul, E. Modiano, and S. Ulukus, "Age of information: An introduction and survey," *IEEE Journal on Selected Areas in Communications*, vol. 39, no. 5, pp. 1183–1210, 2021.
- [7] A. Jaiswal, A. Chattopadhyay, and A. Varma, "Age-of-information minimization via opportunistic sampling by an energy harvesting source," *IEEE Transactions on Cognitive Communications and Networking*, vol. 10, no. 6, pp. 2296–2310, 2024.
- [8] J. Wang, W. Cheng, and H. Vincent Poor, "Statistical delay and error-rate bounded QoS provisioning for AoI-driven 6G satellite-terrestrial integrated networks using FBC," *IEEE Transactions on Wireless Communications*, vol. 23, no. 10, pp. 15 540–15 554, 2024.
- [9] W. Cheng, Y. Xiao, S. Zhang, and J. Wang, "Adaptive finite blocklength for ultra-low latency in wireless communications," *IEEE Transactions on Wireless Communications*, vol. 21, no. 6, pp. 4450–4463, 2022.
- [10] J. Wang, W. Cheng, and W. Zhang, "AI-enabled multi-QoS provisioning for 6G HRLLC," *IEEE Communications Magazine*, pp. 1–7, 2026.
- [11] ITU-R, "Framework and overall objectives of the future development of IMT for 2030 and beyond," *Recommendation ITU-R M.2160-0*, Nov. 2023.
- [12] Z. Sheng, P. Zhu, J. Li, D. Wang, and X. You, "Toward deterministic 6G HRLLC: A multi-agent DRL-based traffic scheduling method in HRLLC and TSN converged networks," *IEEE Transactions on Wireless Communications*, vol. 25, pp. 15 054–15 068, 2026.
- [13] Y. Huang, Y. Jiang, F.-C. Zheng, P. Zhu, and D. Wang, "Energy-efficient resource orchestration for URLLC in cell-free RANs via GNNs with reliability enforcement," *IEEE Transactions on Wireless Communications*, vol. 25, pp. 14 166–14 182, 2026.
- [14] O. Ayan, S. Hirche, A. Ephremides, and W. Kellerer, "Optimal finite horizon scheduling of wireless networked control systems," *IEEE/ACM Transactions on Networking*, vol. 32, no. 2, pp. 927–942, 2024.
- [15] D. Wu and R. Negi, "Effective capacity: a wireless link model for support of quality of service," *IEEE Transactions on Wireless Communications*, vol. 2, no. 4, pp. 630–643, 2003.
- [16] J. Wang, W. Cheng, and W. Zhang, "Performance boundary analyses for statistical multi-QoS framework over 6G SAGINs," *IEEE Transactions on Wireless Communications*, vol. 24, no. 5, pp. 4329–4343, 2025.
- [17] I. Muhammad, H. Alves, N. H. Mahmood, O. L. A. López, and M. Latva-aho, "Mission effective capacity—a novel dependability metric: A study case of multiconnectivity-enabled URLLC for IIoT," *IEEE Transactions on Industrial Informatics*, vol. 18, no. 6, pp. 4180–4188, 2022.
- [18] M. Karbalaee Motaleb, V. Shah-Mansouri, S. Parsaeefard, and O. L. Alcaraz López, "Resource allocation in an open RAN system using network slicing," *IEEE Transactions on Network and Service Management*, vol. 20, no. 1, pp. 471–485, 2023.
- [19] O. Adamuz-Hinojosa, L. Zanzi, V. Sciancalepore, A. Garcia-Saavedra, and X. Costa-Pérez, "ORANUS: Latency-tailored orchestration via stochastic network calculus in 6G O-RAN," in *IEEE INFOCOM 2024 - IEEE Conference on Computer Communications*, 2024, pp. 61–70.
- [20] M. Polese, L. Bonati, S. D'Oro, S. Basagni, and T. Melodia, "CoIO-RAN: Developing machine learning-based xApps for open RAN closed-loop control on programmable experimental platforms," *IEEE Transactions on Mobile Computing*, vol. 22, no. 10, pp. 5787–5800, 2023.
- [21] A. Lacava, M. Bordin, M. Polese, R. Sivaraj, T. Zugno, F. Cuomo, and T. Melodia, "ns-O-RAN: Simulating O-RAN 5G systems in ns-3," in *Proceedings of the 2023 Workshop on ns-3*, 2023, pp. 35–44.
- [22] J. Rao and S. Vrzic, "Packet duplication for URLLC in 5G: Architectural enhancements and performance analysis," *IEEE Network*, vol. 32, no. 2, pp. 32–40, 2018.
- [23] M. Polese, L. Bonati, S. D'Oro, S. Basagni, and T. Melodia, "Understanding O-RAN: Architecture, interfaces, algorithms, security, and research challenges," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 2, pp. 1376–1411, 2023.
- [24] J. Pan, J. Wang, Z. Yun, Z. Xiao, Y. Ouyang, W. Cheng, and W. Zhang, "MAC revivo: Artificial intelligence paves the way," 2024, arXiv:2410.15820.
- [25] 3rd Generation Partnership Project, "Common functional architecture to support mission critical services; stage 2," 3GPP TS 23.280 V19.10.1, May 2026.
- [26] H. S. Wang and N. Moayeri, "Finite-state Markov channel—a useful model for radio communication channels," *IEEE Transactions on Vehicular Technology*, vol. 44, no. 1, pp. 163–171, 1995.
- [27] W. Hoeffding, "Probability inequalities for sums of bounded random variables," *Journal of the American Statistical Association*, vol. 58, no. 301, pp. 13–30, 1963.
- [28] N. Patriciello, S. Lagén, B. Bojović, and L. Giupponi, "An E2E simulator for 5G NR networks," *Simulation Modelling Practice and Theory*, vol. 96, p. 101933, 2019.
- [29] K. Koutlia, B. Bojović, Z. Ali, and S. Lagén, "Calibration of the 5G-LENA system level simulator in 3GPP reference scenarios," *Simulation Modelling Practice and Theory*, vol. 119, p. 102580, 2022.
- [30] J. Luis Herrera, S. Montebugnoli, D. Scotece, L. Foschini, and P. Bellavista, "A tutorial on O-RAN deployment solutions for 5G: From simulation to emulated and real testbeds," *IEEE Communications Surveys & Tutorials*, vol. 28, pp. 1709–1748, 2026.